WEEKLY JOURNAL DIGEST

Weekly journal digest for 2026-08-31 to 2026-09-06

Curated articles in political communication and computational social science.

SPECIAL ISSUE UPDATES

Starting this week, special-issue calls and approaching deadlines will lead the digest. We monitor 36 journals through publisher feeds and add individually verified calls from other journals. Our regular 17-journal article digest continues as before. Coverage varies by journal; the attached coverage note lists the journals and sources.

Calls checked: 2026-09-12.

Manual calls omitted pending renewed official-page verification: Public Relations Review.

Thematic Issue: Revisiting the Nexus of Science, Politics, Media, and Publics Amid Digital and Societal Transformations
Media and Communication | Open | call for papers
Deadline: 2026-09-15
Official call / submission details

Thematic Issue: Trust, Misinformation, and Institutional Communication in High-Trust or High-Control Societies
Media and Communication | Open | call for papers
Deadline: 2026-09-15
Official call / submission details

Special Issue: LLM Sycophancy in Digital Journalism: News Consumption, Public Deliberation, and Governance in the Age of Conversational Agents
Digital Journalism | Open | call for papers
Deadline: 2026-09-30
Official call / submission details

Special Issue: New Directions in Influencer Research
Journal of Advertising Research | Open | call for papers
Deadline: 2026-10-12
Official call / submission details

Special Issue: Older Adult Communication Across Contexts
Communication Quarterly | Open | call for papers
Deadline: 2026-10-15
Official call / submission details

Thematic Issue: Reconceptualizing Deliberative Democracy and Journalism in AI-Mediated Public Spheres
Media and Communication | Opens later | call for papers
Deadline: 2026-10-15
Official call / submission details

Thematic Issue: Social Media Journalism as a One-Way Street?
Media and Communication | Opens later | call for papers
Deadline: 2026-10-15
Official call / submission details

Special Issue: How Communication Technology Mediates Death, Dying, and Grief
Western Journal of Communication | Open | call for papers
Deadline: 2026-10-31
Official call / submission details

Special Issue: Relational Perspectives in Journalism (Studies)
Journalism Studies | Open | call for papers
Deadline: 2026-11-01
Official call / submission details

Special Issue: Conspiracy Rhetoric Twenty-Five Years Later: Conspiring Minds Still Want to Know
Western Journal of Communication | Open | call for papers
Deadline: 2026-11-01
Official call / submission details

Special Issue: The Emotional Turn in Public Relations Theorizing
Journal of Public Relations Research | Open | call for papers
Deadline: 2026-11-06
Official call / submission details

Special Issue: Intergroup Communication
Group Processes & Intergroup Relations | Open | call for papers
Deadline: 2026-12-15
Official call / submission details

Thematic Issue: Children, Platforms, and Power: Rethinking Agency, Harm, and Governance in Datafied Childhoods
Media and Communication | Opens later | call for papers
Deadline: 2026-12-15
Official call / submission details

Thematic Issue: Ecosystems of AI: Synthetic Media, Power, and Disinformation in the Age of Political Simulation
Media and Communication | Opens later | call for papers
Deadline: 2026-12-15
Official call / submission details

Special Issue: Strategic Communication and Metamodernism
International Journal of Strategic Communication | Open | call for papers
Deadline: 2026-12-31
Official call / submission details

Special Issue: Intercultural Communication in the Age of Artificial Intelligence: Negotiating Identity, Culture, and Digital Interaction
Journal of Intercultural Communication Research | Open | call for papers
Deadline: 2026-12-31
Official call / submission details

Special Issue: Reviews, Replications, and Reports
Review of Communication | Open | call for papers
Deadline: 2026-12-31
Official call / submission details

Special Issue: Clarity
Review of Communication | Open | call for papers
Deadline: 2027-01-31
Official call / submission details

Thematic Issue: Mediating Migration: Representation, Inequality, and the Politics of Difference in Contemporary Communication Environments
Media and Communication | Opens later | call for papers
Deadline: 2027-02-15
Official call / submission details

Special Issue: Fast-Forward Communication Theory
Review of Communication | Open | call for papers
Deadline: 2027-02-15
Official call / submission details

Special Collection: Social Group Dynamics
Nature Human Behaviour; Nature Communications; Communications Psychology; Scientific Reports | Open | call for papers
Deadline: 2027-02-18
Official call / submission details

Thematic Issue: ClAImate Change? Environmental Communication, Artificial Intelligence, and the Reordering of Public Life
Media and Communication | Opens later | call for papers
Deadline: 2027-04-15
Official call / submission details

Special Issue: AI and the Future of Advertising Creativity
Journal of Advertising Research | Opens later | call for papers
Deadline: 2027-10-15
Official call / submission details

Special Issue: Artificial Intelligence, Ethics, and Strategic Communication (In partnership with the Arthur W. Page Center for Integrity in Public Communication)
International Journal of Strategic Communication | Open | call for papers
Deadline: 2027-10-31
Official call / submission details

Thematic Issue: Exploring Silence in Public Communication Research
Media and Communication | Opens later | call for papers
Deadline: 2027-11-15
Official call / submission details

Call for Special Issue Proposals
Communication and the Public | Open | special issue proposal
Deadline: Not specified; see official call
Official call / submission details

Standing Call for Special Section and Forum Proposals
International Journal of Communication | Open | special issue proposal
Deadline: Not specified; see official call
Official call / submission details

SUMMARY

This updated COMAP Journal Bot edition completes coverage of August 31-September 6, replacing the earlier collection made before the week ended. It brings together research on how state-backed narratives enter mainstream debate, how election-integrity messages perform in the field, and how AI changes the evidence we can draw from text, images, and simulated respondents.

The methods papers deserve particular attention: a generative-AI framework tackles causal inference with unstructured data, while a large benchmark exposes systematic distortions in digital twins. Catch-up selections examine causal overstatement and declining linguistic diversity in AI-assisted writing. Relevant perspectives and methodological critiques are retained alongside empirical studies; where publishers supplied no abstract, that limitation is stated.

HIGHLIGHTS

State-sponsored agenda setting: Measuring the impact of narrative laundering

PROCEEDINGS OF THE NATIONAL ACADEMY OF SCIENCES · 2026-09-04

Why it matters · Examines narrative laundering in a Russian-aligned campaign, shifting evaluation from direct persuasion to the public agenda shaped by state media, fabricated sites, and paid influencers.

Open article →

Leveraging generative AI for causal inference with unstructured data

PROCEEDINGS OF THE NATIONAL ACADEMY OF SCIENCES · 2026-09-03

Why it matters · Connects causal estimation to text and image representations, with applications to Chinese social-media censorship, political rhetoric, and visual features.

Open article →

Digital twins are funhouse mirrors: Five systematic distortions

SCIENCE ADVANCES · 2026-09-04

Why it matters · Nineteen preregistered studies identify five distortions in simulated respondents, offering a benchmark for deciding when digital twins can support social-science research.

Open article →

Can a public information campaign increase trust in American elections?

SCIENCE ADVANCES · 2026-09-04

Why it matters · A randomized field campaign finds mostly null effects on turnout and broad election confidence, with a narrower gain in vote-by-mail trust-an important check on survey-based persuasion estimates.

Open article →

Measuring Partisanship and Representation in Online Congressional Communications

AMERICAN POLITICAL SCIENCE REVIEW · 2026-09-01

Why it matters · Addresses measurement of partisanship and representation in congressional online communication, directly relevant to research on elite messaging.

Open article →

Beyond beyond standardization: studying robustness of empirical claims based on topic modeling through multiverse analysis

COMMUNICATION METHODS AND MEASURES · 2026-08-31

Why it matters · Uses multiverse analysis to examine robustness of claims based on topic models, making analytical choices part of the evidence assessment.

Open article →

Perceived humanness and empathy are dissociated in relationship advice generated by a large language model

NATURE COMMUNICATIONS · 2026-09-03

Why it matters · Separates perceived humanness from empathy in generated advice, showing how stylistic incentives shape human-AI communication.

Open article →

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- Science Advances
- Nature Human Behaviour
- Nature Communications

Previous Week Catch-Up

- Journal of Communication
- Political Communication
- Communication Research
- American Journal of Political Science
- Science
- Proceedings of the National Academy of Sciences
- Nature Human Behaviour
- Nature Communications
- Nature Machine Intelligence

Late Additions

- Proceedings of the National Academy of Sciences
- Nature Machine Intelligence

COLLECTION SNAPSHOT

TOTAL COLLECTED CANDIDATES	NEW THIS WEEK CANDIDATES	PREVIOUS WEEK CATCH-UP CANDIDATES
414	207	191
LATE ADDITIONS CANDIDATES		
16		

Curated digest below: 28 new, 21 catch-up, 3 late additions

The collected counts are after automatic source filters and before manual relevance review. The full digest and attachments contain the curated selection.

All 17 article sources completed successfully. Catch-up covers August 24-30; late additions were published before August 24.

FULL CURATED DIGEST

New This Week

Journal of Computer-Mediated Communication

A comparative test of the protection motivation theory: active and passive privacy protection in Brazil, Germany, the United States, and Vietnam

2026-09-01 · DOI 10.1093/JCMC/ZMAG019

Yannic Meier; Laurent H Wang; Nicole C Kramer; Miriam J Metzger

Department of Communication, University of California Santa Barbara , Santa Barbara,; Research Center Trustworthy Data Science and Security , Dortmund,; Social Psychology, Media and Communication, University of Duisburg-Essen , Duisburg,; Stan Richards School of Advertising and Public Relations, Moody College of Communication, University of Texas at Austin , Austin,

Abstract. Abstract Online companies surveil users around the world for commercial purposes. We advance protection motivation theory (PMT), by categorizing user responses to surveillance as either active or passive (i.e., self-inhibition) privacy protection behavior, integrating privacy literacy overconfidence, and exploring privacy resignation as a moderator. Finally, we adopt a comparative lens by investigating protection behavior in four national contexts: Brazil, Germany, the United States, and Vietnam. Results of an online survey (Ntotal = 2,092) show that perceiving surveillance as privacy threat was not consistently associated with active protection behavior. Rather, self-inhibition was a more likely response to surveillance which may have detrimental consequences for both individuals and societies. Across countries, perceived effectiveness of active protection positively relates to both forms of protection while active protection does not prevent self-inhibition. Interestingly, privacy resignation may not always impede protection. In summary, not all aspects of PMT are generalizable when using a culturally diverse sample.

Read article →

Context, script, cue: understanding human-AI intimate relationships

2026-09-01 · DOI 10.1093/JCMC/ZMAG016

Fan Yang; Meiling Du; Ningxin Li; Qing Yan

School of Journalism and Communication, Jinan University , Guangzhou, Guangdong,; School of Journalism and Communication, Xiamen University , Xiamen, Fujian,; School of Journalism and Mass Communications, University of South Carolina , Columbia, SC,

Abstract. Abstract The integration of artificial intelligence (AI) into the intimate sphere of human relationships presents a profound challenge to traditional understanding of intimacy, necessitating a critical examination of this technology's extension into a domain that has always been viewed as deeply human. Through in-depth semi-structured interviews (N = 23), this study examines how human-AI intimate relationships develop in Chinese society at the intersection of traditions and modernization. A three-pronged Context-Script-Cue (CSC) framework emerges, capturing how users navigate emotional investment alongside the artificial nature of AI. By empirically examining how users engage in human-AI intimate relationships, this study offers the new CSC framework for understanding the development of human-AI intimate relationships.

Read article →

Political Communication

Democratic Accountability in East Asia: A Comparative Experimental Study of Voter Responses to Norm Violations

2026-08-31 · DOI 10.1080/10584609.2026.2721665

Woo Chang Kang; Sunkyoung Park; Osbern Huang; Koh Iwabuchi; Tetsuro Kobayashi

Griffith University; Korea University; Waseda University

Abstract. Existing research shows that voters in established democracies often fail to penalize politicians who violate democratic norms, as many are willing to trade democratic principles for partisan interests in polarized environments. Yet such partisan double standards remain underexplored in East Asia, where countries share authoritarian legacies, face ongoing risks of democratic backsliding, and operate under the geopolitical shadow of powerful authoritarian neighbors. Using a large-scale vignette experiment conducted in Japan, South Korea, Taiwan, and the United States (N = 10,194), this study examines whether voters sanction politicians who engage in democratic norm violations. We focus on reactions to three types of undemocratic statements: election denial, protest suppression, and attacks on the judiciary. The results show that voters in East Asian democracies impose stronger penalties than their American counterparts, though the degree and pattern of accountability differ across cases. Election denial consistently generates the strongest sanctions, while Japan and Taiwan show robust reactions across all scenarios. South Korea displays comparatively modest responses to protest suppression, and the United States shows the weakest effects overall. These findings highlight both the resilience and the heterogeneity of democratic accountability in East Asia and underscore the need to understand regional variation when assessing the prospects for democratic stability.

Read article →

From #StayWoke to "Culture Wars": How Social Justice Discourse is Separately and Synergistically Politicized on Twitter and YouTube

2026-08-31 · DOI 10.1080/10584609.2026.2717428

Sara Holland Levin; Rui Wang; Junwan Seo; Yi Yin Leong; Emma Bennett; Dongdong Yang; Zhuohui Yang; Matt Chen; Yini Zhang; Jiyoun Suk

University at Buffalo; University of Connecticut

Abstract. This paper explores how the discourse surrounding a racially marginalized community becomes politicized via culture war narratives across platforms. Drawing from identity politics and political polarization research, we discuss how racial justice discourse can be hijacked by partisan opinion leaders and their followers over time and explain how this process plays out through both differentiated and coordinated social media platform use. Situating our discussion in the "woke" discourse as originated within the Black Twitter community, we collect data from 2012 through 2022 on Twitter and YouTube and apply user- and content-level computational analyses. Results show that the term "woke" has been increasingly amplified in culture war conversations on both platforms over time. However, the two platforms exhibited differences: culture war topics dominated on Twitter, while conservative news and commentary channels were most prevalent on YouTube. These two platforms were also interrelated in the politicization process. YouTube was a top domain within external links embedded in tweets, and Twitter was also a top destination within the external links embedded in YouTube video descriptions. Conservatives were especially keen on sharing right-wing content across platforms, from YouTube to Twitter, suggesting the success of cross-platform coordination in amplifying culture wars. Both theoretical and political implications for polarization on social media are discussed.

Read article →

Communication Research

Investigating Conditional Macro-Level Media Effects Over a Long-Term: How Crisis Periods Moderate the Effects of Suicide Coverage on Suicide Rate Change (1861-2007)

2026-09-03 · DOI 10.1177/00936502261480405

Christina Viehmann; Stefan Gei; Manina Biermayer; Oliver Quiring; Florian Arendt

GESIS - Leibniz Institute for the Social Sciences, Cologne, Germany; Johannes Gutenberg University Mainz, Germany; Norwegian University of Science and Technology (NTNU), Trondheim, Norway; University of Vienna, Austria

Abstract. How does media shape society over decades, and does its effect vary depending on the societal atmosphere or zeitgeist ? While theoretically important, there is a lack of systematic empirical evidence on conditional long-term macro-level media effects. We propose a broader framework to study such effects and demonstrate its applicability for the "Werther effect" in the UK. More specifically, we investigate whether media coverage of suicides in a newspaper of record is linked to suicide rates and whether this link changes during societal crises. For this, we combine suicide data with a large-scale automated content analysis of crisis and suicide reporting in The Times over a century (1861-2007). Our analysis shows that increased suicide reporting is associated with higher suicide rates, but only during crisis periods. The results are consistent with a conditional Werther effect and may offer one piece of the larger puzzle of long-term macro-level media influences on society.

Read article →

Experimental Research on Stereotype Expression in Freely Generated Language: A Systematic Literature Review

2026-09-01 · DOI 10.1177/00936502261478981

Kim T. Beenen; Camiel J. Beukeboom; Christian Burgers; Pia Sommerauer

University of Amsterdam, The Netherlands; Vrije Universiteit Amsterdam, The Netherlands

Abstract. Social stereotypes spread through biased language. Experiments in which participants freely generate language allow researchers to causally test how stereotypes affect language use and capture multiple linguistic variables simultaneously, offering high ecological validity. Despite this potential, it remains unclear how and to which extent this paradigm has been used to study linguistic biases. To address this gap, we conducted a systematic literature review (k = 106 experiments) and found that most freely generated language experiments examined only one linguistic variable, leaving the interplay between linguistic variables largely underexplored. To advance our understanding of stereotype expression in everyday language, we recommend experiments to measure multiple linguistic variables simultaneously and in relation to each other. This knowledge is needed to address harmful stereotype communication in the real world through automated detection tools, communication interventions, and the fine-tuning of Large Language Models.

Read article →

Sex, Romance, or Recreation? An Experiment Testing the Effects of Cannabis Brand-generated Social Media Ads on College Students' Expectancies and Intentions to Use Cannabis Prior to Sex

2026-08-31 · DOI 10.1177/00936502261480401

Jessica Fitts Willoughby; Stacey J. T. Hust; Leticia Couto; Jiayu Li; Soojung Kang

DePaul University, Chicago, IL, USA; Washington State University, Pullman, WA, USA

Abstract. Sexual and romantic scripts are used to market cannabis products, now legal for adult-use in several U.S. states. We conducted an online experiment (N = 747) examining the impact of viewing brand-generated social media posts promoting cannabis that included sexual, romantic, and non-sexual non-romantic recreational scripts among college students

in Washington state. Increased perceptions that ads suggested cannabis enhances sex were associated with higher sex-related cannabis expectancies. Among men, wishful identification was associated with lower sexual risk expectancies among those who viewed sexual scripts and a greater likelihood of intending to use cannabis before sex among those who viewed sexual or romantic scripts. Among women, perceiving that ads suggested cannabis enhances sex was associated with a greater likelihood of intending to use cannabis before sex, whereas perceiving that ads suggested cannabis impacts sexual consent was associated with a lower likelihood of intending to use cannabis before sex. Additionally, wishful identification among women was associated with a greater likelihood of intending to use cannabis before sex. Findings highlight associations between perceptions of brand-generated social media posts, expectancies, and intentions to use cannabis before sex among college students and identify mechanisms through which effects may occur.

Read article →

Communication Methods and Measures

Beyond beyond standardization: studying robustness of empirical claims based on topic modeling through multiverse analysis

2026-08-31 · DOI 10.1080/19312458.2026.2714769

Paul Balluff; Christina Viehmann; Maximilian Linde; Yannik Peters; Jun Sun; Chung-Hong Chan

GESIS - Leibniz Institute for the Social Sciences

Abstract. While topic modeling is widely used, some scholars have already announced that the application of topic modeling in social science research is impossible to standardize. Meanwhile, researchers have to make numerous methodological decisions, yet base their empirical claims on just one topic model. This raises the question of robustness about those claims. In this study, we apply the framework of preregistered multiverse analysis to evaluate the robustness of previous empirical claims from six studies that were based on topic modeling. Based on the Open Science materials of these six studies, we slightly modify the original topic modeling procedures, such as preprocessing, number of topic clusters, and topic modeling algorithm, to other defensible choices and determine whether the empirical claims remain consistent. Even though we observe that certain empirical claims are more robust, most claims are distorted considerably by changes to the modeling options. The contributions of this study are twofold. First, we confirm that empirical claims based on just one topic model might not be robust against numerous researcher degrees of freedom. Second, we advocate a wider adoption of preregistered multiverse analysis in social science research for checking the robustness of empirical findings.

Read article →

American Journal of Political Science

The territorial rights of wild animals: Justification

2026-08-31 · DOI 10.1111/AJPS.70087

Pablo P. Castello

Sturm College of Law, University of Denver Denver Colorado USA

Abstract. Abstract Territorial rights (TR) have become foundational in normative political theory, addressing pressing questions about boundary drawing, governance, and resource use. Influential TR theorists Cara Nine and Margaret Moore justify humans' TR based on the functional, social, and identityanchored normative significance of nonsubstitutable places. However, these arguments treat animals as resources, excluding them from normative consideration and political standing. This article analyzes Nine and Moore's arguments to assess whether their justificatory logic applies to wild animals. Using mule deers as a case study, I test whether they have sufficiently weighty interests in occupancy, residency, resource use, and control to ground duties on humans and whether TR theories should continue to treat animals as mere natural resources.

Read article →

American Political Science Review

Are Voters Myopic or Looking for Improvement?

2026-09-01 · DOI 10.1017/S0003055426101853

MARKUS PRIOR; TALBOT M. ANDREWS

Cornell University; Princeton University

Abstract. Abstract Timing of valence outcomes plays an important role in key theories of electoral choice, and different models place different emphases on past and future outcomes. Whereas sanctioning models consider only retrospective evaluations as an accountability mechanism, selection models hold that voters use past and proposed future conditions to infer the incumbent's ability to deliver benefits. Drawing on four surveys run on six different samples, we demonstrate that voters prefer improving conditions in both retrospection and prospection. In retrospection, this pattern is difficult to distinguish from overweighting of temporally proximate conditions, which gave rise to the notion of "the myopic voter." Distinguishing the two processes, we find that people do not ignore early datapoints and reward improving conditions (in both retrospection and prospection). This preference for improvement does not simply reflect an uncritical heuristic for upward trends. People are not universally myopic and many genuinely value improving aggregate conditions retrospectively and prospectively.

Read article →

Measuring Partisanship and Representation in Online Congressional Communications

2026-09-01 · DOI 10.1017/S0003055426101841

MICHAEL KISTNER; MICHAEL HESELTINE; ROBERT ALVAREZ; MAYA FITCH; LUCAS LOTHAMER; ELIZABETH SIMAS

Sam Houston State University; Texas A&M University; University of Houston; University of Oxford; University of Texas at Tyler

Abstract. Social media and the internet have created new ways for representatives to communicate. How have members of Congress responded to these opportunities? We introduce a multi-platform dataset of congressional communications extending back to the onset of the social media era. Using computational language processing, we classify approximately 4.7 million tweets, 2.4 million Facebook posts, and 184,000 email newsletters authored by members of Congress between 2009 and 2022 based on intended purpose, and scale the partisanship of each message along a continuous left-right dimension. After validation, we demonstrate how our data can be used to study partisanship and representation in the contemporary Congress. Importantly, our data show congressional rhetoric has become more partisan and negative as social media usage has increased. We identify one potential mechanism contributing to this trend: partisanship and negativity receive inflated levels of positive engagement on social media relative to other forms of messaging like credit claiming or constituency service.

Read article →

Political Analysis

Correcting for Nonignorable Nonresponse Bias in Ordinal Observational Survey Data

2026-09-04 · DOI 10.1017/PAN.2026.10055

Lukas Laffers; Jozef Michal Mintal; Ivan Sutoris

Matej Bel University in Banska Bystrica; National Bank of Slovakia; Norwegian School of Economics

Abstract. Abstract Many political surveys rely on post-stratification, raking or related weighting adjustments to align respondents with the target population. But when respondents differ from nonrespondents on the outcome itself

(nonignorable nonresponse), these adjustments can fail, introducing bias even into basic descriptives. We provide a practical method that corrects for nonignorable nonresponse by leveraging response-propensity proxies (e.g., a respondent's rating of the interview or interviewer-coded cooperativeness) observed among respondents to extrapolate toward nonrespondents, while directly integrating observable covariates and retaining the benefits of post-stratification with known population shares. The method generalizes the variable-response-propensity framework of Peress (2010) from binary to ordinal outcomes, which are widely used to measure trust, satisfaction and policy attitudes. The resulting estimator is computed by maximum likelihood and implemented in a compact R routine that handles both ordinal and binary outcomes. Using the 2024 American National Election Study, we show that accounting for nonignorable nonresponse produces substantively meaningful shifts for life satisfaction (estimated latent correlation 0.47 $\rho \approx 0.47$ rho almost equals 0.47), while yielding only modest changes for retrospective economic evaluations (0.14 $\rho \approx 0.14$ rho almost equals 0.14), highlighting when nonignorable nonresponse substantively affects survey estimates.

Read article →

Nature

Robust inference and correlates from genetic associations with personality

2026-09-02 · DOI 10.1038/S41586-026-10992-9

Ted Schwaba; Margaret L. Clapp Sullivan; Wonuola A. Akingbuwa; Kerli Ilves; Peter T. Tanksley; Camille M. Williams; Yavor Dragostinov; Travis T. Mallard; Justin D. Tubbs; Wangjingyi Liao; Lindsay S. Ackerman; Josephine C. M. Fealy; Gibran Hemani; Javier de la Fuente; George Davey Smith; Priya Gupta; Murray B. Stein; Joel Gelernter; Daniel F. Levey; Urmo Vosa; Liisi Ausmees; Anu Realo; Tonu Esko; Mariliis Vaht; Juri Allik; Tonu Esko; Rene Mottus; Uku Vainik; Gudrun A. Jonsdottir; Gudmar Thorleifsson; Arni Freyr Gunnarsson; Gyda Bjornsdottir; Thorgeir E. Thorgeirsson; Hreinn Stefansson; Kari Stefansson; Rosa Cheesman; Qi Qin; Elizabeth C. Corfield; Helga Ask; Fartein Ask Torvik; Eivind Ystrom; Martin Tesli; Dorret I. Boomsma; Eco J. C. de Geus; Jouke-Jan Hottenga; Dener Cardoso Melo; Harold Snieder; Catharina A. Hartman; Charley Xia; Archie Campbell; Michelle Luciano; Ian J. Deary; W. David Hill; Seon-Kyeong Jang; Scott I. Vrieze; Goncalo Abecasis; Michelle K. Lupton; Brittany L. Mitchell; Petra V. Viher; Lucia Colodro-Conde; Nicholas G. Martin; Sarah E. Medland; Eske M. Derks; Briar Wormington; Jaakko Kaprio; Karri Silventoinen; Teemu Palviainen; Agnieszka Musial; Kaili Rimfeld; Robert Plomin; Margherita Malanchini; Danielle M. Dick; Fazil Aliev; Laura W. Wesseldijk; Fredrik Ullen; Miriam A. Mosing; Henry R. Kranzler; Yaira Nunez; Sarah Beck; Renato Polimanti; Tobias Edwards; Alexandros Giannelis; Emily A. Willoughby; James J. Lee; Matt McGue; Antonio Terracciano; Michele Marongiu; Edoardo Fiorillo; Francesco Cucca; Angelina R. Sutin; Peter J. van der Most; Albertine J. Oldehinkel; Tina Kretschmer; Andrey A. Shabalin; Anna R. Docherty; Robert F. Krueger; Colin D. Freilich; Binisha H. Mishra; Terho Lehtimaki; Olli T. Raitakari; Mika Kahonen; Aino Saarinen; Henrik Dobewall; Liisa Keltikangas-Jarvinen; Klaus Berger; Marisol Herrera-Rivero; Fabian Streit; Swapnil Awasthi; Stephanie H. Witt; Johanna Tuhkanen; Katri Raikkonen; Johan G. Eriksson; Jari Lahti; Gail Davies; Paul Redmond; Adele Taylor; Janie Corley; Tom C. Russ; Marina Ciullo; Teresa Nutile; Jun Ding; Yong Qian; Toshiko Tanaka; Luigi Ferrucci; Lea Zillich; Lea Sirignano; K. Paige Harden; Erhan Genc; Patrick D. Gajewski; Stephan Getzmann; Christoph Fraenz; Javier E. Schneider Penate; Stefanie Lis; Alisha S. M. Hall; Christian Schmahl; Sabine C. Herpertz; Abdel Abdellaoui; Michel G. Nivard; Elliot M. Tucker-Drob

Abstract. Personality traits describe stable differences in how people think, feel and behave, and how they interact with and experience their social and physical environments1,2. Many questions remain unanswered about associations between DNA and personality traits, such as their robustness, their generalizability and the biological and social pathways through which they act. Here we meta-analyse data across 46 cohorts comprising 611,037 to 1.14 million participants with European-like and African-like genomes for genome-wide association studies (GWAS) of the Big Five personality traits (extraversion, agreeableness, conscientiousness, neuroticism and openness to experience), and data from up to 50,725 participants for within-family GWAS. We identify 1,260 lead genetic variants associated with personality, including 824 novel variants3. Common genetic variants explain a moderate 4.8-9.3% of the variance in measures of each trait, and 9.3-13.3% among instruments with typical measurement reliability. Genetic associations with personality are highly consistent but not identical across geography, reporter (self versus close other), age group and measurement instrument, and we find minimal spousal assortment for personality in recent history. In contrast to many other social and behavioural traits4,5, within-family GWAS and polygenic index analyses indicate that genetic associations with personality are minimally confounded by the shared family environment. Polygenic prediction, genetic correlation and Mendelian randomization analyses indicate that personality

traits have widespread, potentially causal associations with consequential behaviours and life outcomes. Overall, we find that the genetic architecture of personality is robustly generalizable, minimally confounded and widely relevant to human experience. The genetic architecture of personality is robustly generalizable, minimally confounded and widely relevant to human experience.

Read article →

Proceedings of the National Academy of Sciences

State-sponsored agenda setting: Measuring the impact of narrative laundering

2026-09-04 · DOI 10.1073/PNAS.2534920123

Patrick L. Warren; Darren L. Linvill; Camille J. Saucier

Department of Communication, Clemson University; John E. Walker Department of Economics, Clemson University; Media Forensics Hub, Watt Family Innovation Center, Clemson University

Abstract. Foreign information manipulation and interference (FIMI) is an important part of the digital information ecosystem, yet its effects are poorly understood. Most research analyzing FIMI to date is descriptive, exposing efforts to erode trust, promote favorable narratives, and target specific communities. Existing work evaluating the impacts of FIMI contrasts those directly exposed to FIMI messaging (by viewing the original messages) with those who were not, finding little evidence of changes in beliefs or behavior. Rather than persuading individuals directly, we argue FIMI often operates through agenda-setting processes that shape which issues and narratives dominate public discourse. A key component of this process is narrative laundering, whereby misleading content is scrubbed of its origins and circulated into mainstream channels. This study examines the Russian-aligned Storm-1516 campaign, which uses state media, fabricated sites, and paid influencers to elevate false narratives, especially those that implicate Ukrainian President Volodymyr Zelenskyy in corrupt acts. By analyzing the dissemination of these narratives, we assess how Storm-1516 shifts the public agenda, illustrating how foreign influence can exert substantial effects on information environments even when individual-level persuasive effects may be limited.

Read article →

Leveraging generative AI for causal inference with unstructured data

2026-09-03 · DOI 10.1073/PNAS.2530532123

Kosuke Imai; Kentaro Nakamura

Department of Government, Harvard University; Department of Statistics, Harvard University; John F. Kennedy School of Government, Harvard University

Abstract. We introduce GenAI-Powered Inference (GPI), a statistical framework for causal inference using unstructured data, including text and images. GPI leverages open-source pretrained Generative AI (GenAI) models-such as large language models and diffusion models-not only to generate unstructured data at scale but also to extract low-dimensional representations that are guaranteed to capture their underlying structure. Applying machine learning to these representations, GPI enables estimation of causal effects while quantifying estimation uncertainty. Unlike existing approaches to representation learning, GPI does not require fine-tuning of GenAI models, making it computationally efficient and broadly accessible. We illustrate the versatility of the GPI framework through three applications: 1) estimating the effects of Chinese social media censorship while adjusting for textual confounders, 2) isolating the impact of specific image features from that of other correlated features in the same image, and 3) assessing the persuasiveness of political rhetoric. An open-source software package is available for implementing GPI.

Read article →

A three-decade-long analysis shows rising threatened species and responsibility gaps in global supply chains

2026-08-31 · DOI 10.1073/PNAS.2604868123

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Abstract. Rapid globalization and its accordant supply chains accelerates biodiversity loss. Yet timely assessments of biodiversity status and trends remain insufficient, limiting effective conservation actions in response to supply chain dynamics. Here, we provide a three-decade-long analysis of status and trends of three main biodiversity threats and 21,207 threatened species along production, trade, and final consumption between 185 countries. Results show escalating trends and newly identified threat-species-country-activity networks in biodiversity impacts across global supply chains. We identify changes in people's consumption structure and species intensity of economic activities as the main drivers of this escalation. International trade contributes a large and increasing proportion of total threatened species in most countries, especially for imports to the United States, Japan, and exports from Madagascar, Papua New Guinea. In particular, the bilateral trade between the United States and Madagascar causes the largest number of threatened species (191 17) among all bilateral trade flows worldwide. Furthermore, a unique pattern is revealed from our biodiversity assessment, unlike other environmental pressures such as carbon footprints, that species threatened by per-capita final demand in Latin America, and Caribbean (a developing region) is at least 38% higher than those in developed regions and three times the global average. A timely understanding of shared responsibilities of producer- and consumer-nations for biodiversity threats and status offers a scientific basis for fair and accountable biodiversity governance.

Read article →

Aligning private incentives with societal goals in smallholder agriculture

2026-08-31 · DOI 10.1073/PNAS.2536548123

Gashaw T. Abate; Kaleab Baye; Tanguy Bernard; Jordan Chamberlin; Ruth Hill; Rachid Moussadek; Lamine Senghor

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Abstract. Small-scale farmers in low-income countries are often encouraged to adopt new inputs and practices that generate significant societal benefits, such as improved nutrition, food safety, and environmental sustainability. The limited observability of these attributes, combined with small production scale and misaligned private incentives, leads to adoption levels below the social optimum. Addressing this issue often requires combining instruments identified by different scientific disciplines, spanning across multiple policy sectors, an exercise that is rarely articulated jointly. This paper aims to clarify this, through a simple conceptual framework that explains how farmers decide whether to adopt socially desirable inputs and practices, given the returns and costs they face, their access to financial instruments, and the information and preferences that shape their decisions. It identifies the relevant policy channels, and asks whether the social returns justify the public cost of delivering them. We apply it to three case studies: aflatoxin mitigation, biofortified crop varieties, and conservation agriculture. For each, we identify which parameters bind, which instruments can move them, and how emerging technologies may further contribute. The agenda that follows moves from seeking single-instrument fixes toward identifying the right combination of existing and emerging policy-actionable tools.

Read article →

Cousin marriage and reduced exposure to intimate partner violence and coercive control

2026-08-31 · DOI 10.1073/PNAS.2600906123

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Abstract. Cousin marriage, practiced by over 10% of the world's population, restructures kinship networks by overlapping blood and affinal ties. Theory makes competing predictions about how such kin density shapes intrahousehold conflict. "Protection" accounts posit that consanguinity aligns spouses' interests and increases kin oversight, reducing men's incentives and opportunities to use coercion. "Constraint" accounts, by contrast, emphasize that dense kin networks can prioritize family cohesion over women's autonomy, potentially suppressing help-seeking and facilitating control. Here, I adjudicate between these hypotheses using data on ~36 to 46,000 women from Egypt, Jordan, Pakistan, and Turkiye. Using cluster-fixed-effects models that compare women within the same Demographic and Health Survey primary sampling units and survey waves, I find that marriage to a first cousin is associated with a 2.5 percentage-point lower probability of physical intimate partner violence relative to nonconsanguineous unions (11% lower relative risk). Parallel reductions are observed for coercive controlling behaviors and estimates are robust to matching on observed confounders. The protective association for physical violence is stronger for patrilateral than matrilateral cousins, consistent with protection depending on kin with recognized authority over men in patrilineal systems rather than generic relatedness. In Turkiye, the association persists after adjusting for both spouses' parental consanguinity, suggesting that it is not explained solely by family-level characteristics associated with the intergenerational practice of cousin marriage. These findings indicate that, in these settings, consanguinity is associated with lower reported violence and control.

Read article →

Science Advances

Can a public information campaign increase trust in American elections?

2026-09-04 · DOI 10.1126/SCIADV.AEC8703

Thad Kousser; Seth J. Hill; Neil Malhotra; Robert M. Stein

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Abstract. Confidence in the conduct and results of American elections has declined and polarized along party lines in recent years. Can a nonpartisan public information campaign featuring election officials enhance trust in elections when delivered to skeptical voters during a contentious political contest? We analyze a preregistered, randomized field experiment conducted during the 2024 US presidential election in which conservative voters were randomly assigned to a control condition or to receive a multimodal messaging campaign designed by political professionals explaining protections on election integrity. In contrast to the large positive effects of informational messages found in recent survey experiments, we observed primarily null effects with respect to turnout (n = 147,488), general confidence in elections (n = 4740 to 4868), and perceptions of the frequency of fraud (n = 4925). We observed one hypothesized effect: an increase in trust in the vote-by-mail process (n = 6438). This study provides rigorous evidence from a real public information campaign on election integrity, yielding effect sizes in the field that are much smaller than those observed in prior survey settings.

Read article →

Digital twins are funhouse mirrors: Five systematic distortions

2026-09-04 · DOI 10.1126/SCIADV.AEH8260

Tianyi Peng; Melanie Brucks; George Gui; Daniel J. Merlau; Grace Jiarui Fan; Malek Ben Sliman; Eric J. Johnson; Abdullah Althenayyan; Silvia Bellezza; Dante Donati; Hortense Fong; Elizabeth Friedman; Ariana Guevara; Mohamed Hussein; Kinshuk Jerath; Bruce Kogut; Akshit Kumar; Kristen Lane; Hannah Li; Vicki Morwitz; Oded Netzer; Patryk Perkowski; Olivier Toubia

Columbia University, New York, NY 10027, USA.; Yale University, New Haven, CT 06511, USA.; Yeshiva University, New York, NY 10033, USA.

Abstract. Scientists and practitioners are aggressively moving to deploy digital twins-large language model (LLM)-based models of individuals-across social science and policy research. We conducted 19 preregistered studies with 164 diverse outcomes (e.g., attitudes toward hiring algorithms and intention to share misinformation) and compared human responses with those of their digital twins (trained on each person's previous answers to more than 500 questions). We establish an empirical benchmark for digital twin performance: Digital twins' answers are only modestly more accurate than those from the (homogeneous) base LLM and correlate weakly with human responses (average correlation coefficient of 0.20). To guide future development, we document five ways in which digital twins distort human behavior: (i) insufficient individuation, (ii) stereotyping, (iii) representation bias, (iv) ideological biases, and (v) hyper-rationality. We make our full dataset and code public as a standardized testbed. Our results caution against premature deployment while laying the groundwork for the transparent, replicable, and iterative science necessary for responsible deployment of digital twins.

Read article →

Nature Human Behaviour

Gender and LGBTQ+ representation across college majors and achievement levels using nationally representative US data

2026-09-03 · DOI 10.1038/S41562-026-02510-8

Joseph R. Cimpian; Zachary T. McDermott; Jo R. King; Nathaniel Woznicki; Taek H. Kim

Abstract. Abstract unavailable.

Read article →

The gender gap reconsidered

2026-09-03 · DOI 10.1038/S41562-026-02448-X

Bryce E. Hughes

Abstract. Abstract unavailable.

Read article →

Saturation of moral language predicts lower content engagement on social media

2026-09-02 · DOI 10.1038/S41562-026-02560-Y

Cristian Candia; Mohammad Atari; Nour Kteily; Brian Uzzi

Abstract. Abstract unavailable.

Read article →

Turning biometrics into a commodity raises privacy and governance risks

2026-09-02 · DOI 10.1038/S41562-026-02587-1

Angel Judel; Jose M. Munoz; Carlos Amunategui Perello; Diego Borbon; Jose Antonio Iglesias; Jose Angel Marinaro; Natalia Monti; Moises Sanchez; Ciro Colombara; Amanda de la Fuente; Aldo Diaz

Abstract. Abstract unavailable.

Read article →

Nature Communications

Perceived humanness and empathy are dissociated in relationship advice generated by a large language model

2026-09-03 · DOI 10.1038/S41467-026-77350-1

Bennett Kleinberg; Jari Zegers; Jonas Festor; Stefana Vida; Julian Prasent; Riccardo Loconte; Sanne Peereboom

Abstract. Differentiating generated and human-written content is increasingly difficult. We examine how incentives to convey humanness and task characteristics shape perceptions of textual content across five studies. In Studies 1-2, humans and GPT-4 write relationship advice or relationship descriptions with or without instructions to appear human. New participants judge each text's source. Instructions to appear human increase perceived humanness only for GPT-4, reducing the human advantage. Study 3 shows that these effects persist under instructions to avoid sounding like an AI model and replicate across GPT-4 and GPT-4o. Study 4 demonstrates that GPT models can produce empathy without humanness and humanness without empathy. Study 5 shows that GPT-4 increases perceived humanness by adopting informal, conversational language and other surface-level stylistic cues, whereas human writers show little adjustment. Together, the findings indicate that perceived humanness depends on incentives to appear human and the context in which writing is produced. Large language models can generate empathic language similar to human writing. The authors show that empathic language and perceived humanness are dissociated, revealing stochastic empathy in AI systems.

Read article →

Toward an integrated and globally equitable understanding of political polarization

2026-09-02 · DOI 10.1038/S41467-026-76110-5

Max Falkenberg; Matteo Cinelli; Alessandro Galeazzi; Christopher A. Bail; Rosa M. Benito; Axel Bruns; Anatoliy Gruzd; David Lazer; Jae Kook Lee; Jennifer McCoy; Kikuko Nagayoshi; David G. Rand; Antonio Scala; Alexandra A. Siegel; Sander van der Linden; Onur Varol; Ingmar Weber; Magdalena Wojcieszak; Fabiana Zollo; Andrea Baronchelli; Walter Quattrociocchi

Abstract. Political polarization poses major societal risks, yet a globally relevant, interdisciplinary understanding remains lacking, as nearly half of all studies focus on the United States. We argue for renewed effort to bring global equity to polarization research, highlighting insights robust across countries, those unique to specific contexts, and key literature gaps. Closing these gaps means overcoming cultural and systemic barriers, including data-access restrictions and misaligned incentives. It also demands interdisciplinarity bridging traditional approaches and those addressing the polarizing role of the internet, social media, and AI. Otherwise, efforts to counter polarization, and its democratic harms, risk resting on unsuitable evidence.

Read article →

Unveiling hidden air pollution exposure and impacts with low-cost sensor network-based frameworks

2026-09-02 · DOI 10.1038/S41467-026-77148-1

Navdeep Agrawal; Nimit Godhani; Sourangsu Chowdhury; P. C. Anandh; Anil Kumar; Piyush Rai; Sachchida Nand Tripathi

Abstract. India has made some progress in mitigating severe air pollution; however, many regions continue to experience fine particulate matter (PM2.5) levels exceeding World Health Organization and national standards. Despite this, local-scale exposure information beyond urban centers, intra-regional inequities, and short-term mortality impacts remain poorly understood. This study integrates dense, low-cost sensor-based PM2.5 measurements with machine learning models, recent

family health survey data, and India-specific relative risk information to assess sub-regional exposure inequity among socioeconomic groups and short-term mortality impacts in Bihar, one of the most polluted and understudied states of India. Here we show a clear north-south divide in PM2.5 exposure, with northern districts having mean exposure levels 1.2 times higher. District-level mortality (per 100,000) varied up to 1.7-fold, with daily mortality showing 3.2-fold variability. We observe limited exposure inequity at the state level but substantial inter- and intra-regional inequities within Bihar. This study integrates dense, low-cost sensor-based PM2.5 measurements with machine learning models, recent family health survey data, and India-specific relative risk information to assess sub-regional exposure inequity among socioeconomic groups and short-term mortality impacts in Bihar, one of the most polluted and understudied states of India.

Read article →

The changing surface of the world's roads

2026-09-01 · DOI 10.1038/S41467-026-76234-8

Sukanya Randhawa; Guntaj Randhawa; Clemens Langer; Francis Andorful; Benjamin Herfort; Daniel Kwakye; Omer Olchik; Sven Lautenbach; Alexander Zipf

Abstract. Abstract Resilient road infrastructure is a cornerstone of the UN Sustainable Development Goals (SDGs), yet spatially consistent information on road surface conditions at the global scale remains limited. We overcome this gap based on a deep learning framework that extracts both road pavedness and road width from Planetscope satellite imagery. Thereby, mapping 95.5% of 9.2 million km of critical arterial roads for 2020 and 2024, nearly half of which was previously unclassified.The resulting dataset and multi-scale analytical framework enable assessments of road infrastructure in the context of development, climate adaptation, and equity.Unlike proxies such as nighttime lights, the data directly captures the physical infrastructure underpinning resilience at higher spatial resolution.This dataset reveals a multi-scale geography of human development, capturing infrastructure investment and accessibility at the planetary scale, enabling a global Humanitarian Passability Matrix for humanitarian logistics, while also providing local insights into road disparities-for example, revealing governance-related patterns in Ghana and highlighting climate-related vulnerabilities in Pakistan.

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Previous Week Catch-Up

Journal of Communication

Do good chat groups make good neighbors? Continuity and change of norms in neighbor relations under digital co-presence

2026-08-30 · DOI 10.1093/JOC/JQAG013

Emilija Gagrcin; Olga Pasitselska

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Abstract. Abstract Based on interviews with urban residents in Germany and Israel, this study examines how residential instant messaging groups (e.g., WhatsApp, Telegram) reshape the norms of "good neighboring": reciprocity, distance, and speaking out. Drawing on Blokland's public familiarity framework and Rosenblum's theorizing of neighborly norms, we ask how digital co-presence transforms the relational conditions of neighboring and how these norms are negotiated within residential chat groups. We show that messenger groups lower the threshold for mutual support but introduce tensions through persistent mutual awareness, which are thenresolved through norm negotiation. Extending the public familiarity framework, we identify a complementary axis of performativity-discretion that manifests through norm negotiation. Residents calibrate between engagement and restraint, enacting "good enough" neighboring within the double-bounded constraints of a

building and a chat group. We contribute to research on digitally mediated neighborhood communication by showing how urban dwellers achieve pragmatic coexistence through calibrated visibility.

Read article →

Political Communication

Social Media Savvy? The Use and Effectiveness of Political Parties' Issue Strategies Across Platforms in the German Federal Election Campaign 2025

2026-08-26 · DOI 10.1080/10584609.2026.2715702

Christoph Ivanusch; Simon M. Luebke; Anna-Katharina Wurst; Jorg Haler

Leipzig University; Ludwig-Maximilians-Universitat Munchen; University of Oslo

Abstract. How does the ever-growing relevance of social media shape political campaigning? A vibrant public and academic debate addresses the role of social media in elections and democratic processes. Yet, despite a surge in research, we still know little about the extent to which political parties adjust established issue strategies to the priorities of different social media platforms' audiences and how effective these are in generating user engagement. Drawing on theories of issue competition - issue ownership, riding-the-wave, and issue yield - we analyze parties' social media communication during the 2025 German federal election campaign on Facebook, Instagram, and TikTok. Combining manual quantitative content analysis of party and candidate posts with large-scale survey data from the German Longitudinal Election Study (GLES), we assess both strategic use and effectiveness of issue strategies. Our results show that parties strategically prioritize owned issues and those salient on the general public agenda, often combining both dimensions in issue yield strategies. However, these strategies only partly align with what the specific platform audiences value and reward. User engagement is highest for posts that match platform-specific issue preferences, especially when the issues are salient both to a party's supporters on that platform and to its wider user base. The findings offer new insights into the dynamics of contemporary digital campaigning and cross-platform party competition.

Read article →

Communication Research

Less Effort to Believe and More Effort to Disbelieve: Examining the Interplay between News-Finds-Me and Affective Polarization in Political Fake News Susceptibility in the United States

2026-08-26 · DOI 10.1177/00936502261472023

Yu Tian; Lars Willnat; Baohua Zhou

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Abstract. Drawing upon an online survey of 719 US adult partisans incorporating response latency data, we connected News-Finds-Me (NFM) to the partisan-motivated reasoning literature and theorized it as a consensus-based mode of news exposure that leads to different news-processing pathways, where the vested news-processing effort depends on whether the political falsehood aligns with one's partisan identity and surrounding majority opinion. Individual-level analysis and headline-level modeling jointly illuminated the asymmetric interplay between NFM and affective polarization. In partisan-consistent scenarios, high-NFM users devoted limited processing time, falling short of cognitive effort to enable affect-laden, directional believing of partisan-consistent fake news. However, when challenged by partisan inconsistency, high-NFM users dedicated significantly more processing time, which exacerbated the relationship between affective polarization and refutation of partisan-inconsistent falsehoods. Contrarily, such an effort discrepancy, widely documented in the partisan-motivated reasoning literature, was weakened or reversed for low-NFM individuals. Extending the "Effort-Reinforces-Directional-Reasoning" thesis to fake news vulnerability, our findings underscore the paradoxical interplay between NFM and affective polarization: while NFM and affective polarization deviate people from active news-seeking and

accuracy-oriented reasoning, their joint dynamics form a type of group thinking that reduces vulnerability to partisan falsehoods-even those aligned with one's partisan attachments.

Read article →

From Exposure to Creation: Cultivating Narrative Ownership to Reduce Stigma

2026-08-24 · DOI 10.1177/00936502261478972

Xun Zhu

University of Kentucky, Lexington, USA

Abstract. Stories are powerful agents of attitude change, including attitudes toward stigmatized groups. However, existing theories have focused on narrative exposure, with limited attention to narrative creation. Creating a story involves simulating characters and events and expressing them through language, which may engage unique psychological processes. This research investigates how and why creating stories shapes stigma. In three experiments, participants wrote, imagined, or read a positive story about a stigmatized character. Study 1 showed that narrative creation, especially writing, increased psychological ownership, which predicted greater transportation and lower stigma. Study 2 replicated these findings using a different stigmatized group. Study 3 showed that psychological ownership continued to explain the effects of narrative creation above and beyond perceived agency. These findings extend theories of narrative persuasion by considering the effects of story creation on intergroup attitudes and introducing psychological ownership as a novel mechanism linking creation to narrative processing and outcomes.

Read article →

American Journal of Political Science

Partisan bias in a sorted party system: A problem of epistemic inequality

2026-08-26 · DOI 10.1111/AJPS.70086

Justin Pottle

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Abstract. Abstract Normative and empirical scholars typically treat partisans' distrust of their political rivals as a product of individual cognitive biases. But even if bias is inevitable, it is parties' coalitionbuilding and messaging strategies that determine who bias is directed toward. This paper argues that the social sorting of party conflict exacerbates the epistemic and democratic harm of partisan distrust by generating distinctive patterns of epistemic inequality. I show that when voters associate parties with particular social groups, members of those groups suffer greater risk of testimonial injustice and related credibility deficits owing to their perceived partisanship. These dynamics are most damaging for members of marginalized groups, as party-group overlap can reinforce the prejudices they already face. Addressing these harms, I argue, requires looking beyond individual partisans to the background features of party systems that shape how elites mobilize group identities and animus in the struggle for office.

Read article →

Science

Correcting the logic behind the 2026 US vehicle emissions standard rollback

2026-08-27 · DOI 10.1126/SCIENCE.AEF0464

Antonio M. Bento; Kenneth T. Gillingham; Mark R. Jacobsen; Christopher R. Knittel; Benjamin Leard; Joshua Linn; David S. Rapson; James M. Sallee; Arthur A. van Benthem; Kate S. Whitefoot

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Abstract. Correction of selective assumptions and inconsistent interpretations of evidence would overturn policy justification

Read article →

The worst energy policy in the world

2026-08-27 · DOI 10.1126/SCIENCE.AEJ2018

Paasha Mahdavi; Michael L. Ross

Michael L. Ross is a professor in the Department of Political Science and the Institute of the Environment and Sustainability at the University of California, Los Angeles, Los Angeles, CA, USA.; Paasha Mahdavi is an associate professor in the Department of Political Science and (by courtesy) the Bren School of Environmental Science and Management at the University of California, Santa Barbara, Santa Barbara, CA, USA.

Abstract. The Iran war could serve as a turning point in accelerating the global transition away from fossil fuels: Every additional day that oil and gas prices are high and future supplies are uncertain, wind and solar power-cheaper and more scalable than ever before-grow all the more appealing. But nearly half of the world's countries have adopted emergency subsidies that research shows will make this transition slower, more expensive, and less equitable.

Read article →

Proceedings of the National Academy of Sciences

Perfectibility in science and democracy: Institutions that learn

2026-08-26 · DOI 10.1073/PNAS.2626408123

Harvey V. Fineberg

Department of Health Policy and Management, Emeritus, Harvard T.H. Chan School of Public Health

Abstract. Abstract unavailable.

Read article →

Sensory context improves language prediction in humans and LLMs

2026-08-26 · DOI 10.1073/PNAS.2600317123

Thomas L. Botch; Emily S. Finn

Department of Psychological and Brain Sciences, Dartmouth College

Abstract. Language is a fundamental human capacity. Large language models (LLMs) have presented the first viable model of language outside of humans, yet how these models learn and use language differs significantly from humans. Here, we compare LLMs and humans predicting language with varying levels of sensory information-from disembodied written text to audiovisual videos of speakers-to demonstrate that, in both humans and LLMs, sensory context is critical for optimal performance. We asked human participants to predict upcoming words within narratives presented as either audiovisual, audio-only, or written language (N = 1 , 500 total; 500 per modality), focusing on content words that convey essential meaning. We compared these predictions across modalities as well as to predictions generated by LLMs. Human predictions were overall more accurate than LLMs, regardless of the sensory modality of presentation. Compared to written language, both audiovisual and audio-only language increased the accuracy and consensus of human predictions and decreased

alignment with LLMs. We identified that prosody, a signal that conveys important linguistic information via the auditory channel, partially drove both the observed advantage in accuracy within humans and divergence from LLMs. Integrating multimodal information-i.e., prosody, auditory, or audiovisual information-with the representation of language learned during LLM training improved models' next-word prediction performance and increased the efficiency of language learning. These findings demonstrate that sensory contexts are foundational to human-like language behavior, and that these contexts can enrich and accelerate language acquisition within LLMs similar to what is observed within human development.

Read article →

Interactive representations of power and status signals in learning social hierarchies

2026-08-25 · DOI 10.1073/PNAS.2532908123

Chunliang Feng; Feilong Liu; Binjie Yang; Edmund Derrington; Yuejia Luo; Chen Qu; Jean-Claude Dreher

Center for Studies of Psychological Application, South China Normal University; Faculty of Health and Wellness, City University of Macau; Guangdong Key Laboratory of Mental Health and Cognitive Science, South China Normal University; Institute for Neuropsychological Rehabilitation, University of Health and Rehabilitation Sciences; Key Laboratory of Brain, Cognition and Education Sciences (South China Normal University), Ministry of Education; Neuroeconomics, Reward and Decision-Making Team, Institut des Sciences Cognitives Marc Jeannerod, CNRS; Philosophy and Social Science Laboratory of Reading and Development in Children and Adolescents (South China Normal University), Ministry of Education; School of Psychology, Shenzhen University; School of Psychology, South China Normal University; The State Key Lab of Cognitive and Learning, Faculty of Psychology, Beijing Normal University; University Claude Bernard Lyon

Abstract. Social hierarchies, which organize our social relationships, have heterogeneous effects on choice behavior. Two central dimensions of social hierarchies, status, which is earned through recognition of competence, and power, which is the capacity to control the outcomes of others, may contribute to different behavioral profiles. Yet, the neurocomputational processes by which status- and power-related signals are learned based on social feedback, and influence subsequent decision-making, remain unclear. Here, we combine functional neuroimaging with computational modeling to examine how individuals acquire knowledge about others' status and power, and how these learned hierarchy signals shape social influence on advice-taking during a risky choice task. We found that the medial prefrontal cortex covaried with status estimates, whereas the amygdala covaried with power estimates. Moreover, estimates in one dimension influenced choices in the other during learning. This indicates interactions between status and power signals, with retrosplenial cortex activity associated with hierarchy signals beyond immediate choice demands. Finally, status broadly modulated response biases toward both safe and risky advice, whereas power primarily increased bias toward safe advice. These findings characterize how experimentally instantiated status and power signals are learned and influence risky decision-making, indicating partially separable yet interactive representations of these hierarchy signals.

Read article →

Nature Human Behaviour

Mentalizing and heuristics as distinct cognitive strategies in human teaching

2026-08-26 · DOI 10.1038/S41562-026-02540-2

Sevan K. Harootonian; Thomas L. Griffiths; Yael Niv; Mark K. Ho

Abstract. Teaching is a foundational social behaviour that can result from mentally effortful reasoning or cognitively frugal heuristics. When do people use these strategies to teach? Here we investigated this using behavioural experiments and computational modelling with adult participants recruited via Prolific. Experiment 1 (N = 100) revealed robust individual differences: some participants taught by reasoning about a learner's knowledge, consistent with an optimal Bayesian pedagogy model, while others relied on simple heuristics that do not require mentalizing. In two preregistered follow-up experiments, we found that people persist in using heuristics even when they are no longer effective (experiment 2, N = 253, P < 0.001, rank-biserial r = 0.287, 95% confidence interval 0.149-0.419) but that this tendency is pre-empted when inference about a learner's knowledge is scaffolded using an auxiliary task (experiment 3, N = 759, P < 0.001, partial $${\eta }_{{\rm{p}}}^{2}=0.107$$, 95% confidence interval 0.068-0.148). These results demonstrate sophisticated arbitration between

planning and heuristics during teaching and elucidate the more general mechanisms involved in adapting mental effort during social interactions. Harootonian et al. find that people switch between effortful, tailored teaching and easier, simple shortcuts, favouring shortcuts even when unhelpful. Prompting participants to reflect on learners' limitations restores effective, personalized instruction.

Read article →

Public perceptions of polygenic testing and embryo selection for non-medical traits

2026-08-26 · DOI 10.1038/S41562-026-02562-W

Edmond Awad; Clara Colombatto; Joanna Demaree-Cotton; Brian D. Earp; Jim A. C. Everett; Peng Liu; G. Owen Schaefer; Ilina Singh; Dominic Wilkinson; Julian Savulescu

Abstract. Abstract Some advances in reproductive technologies raise substantial ethical and psychological challenges, for example, the use of preimplantation genetic testing to select embryos based on non-medical traits. While studies have explored public willingness to use preimplantation genetic testing for medical or non-medical attributes, stated willingness may not reflect implantation decisions when such information is available. This large cross-national study examined public views on polygenic embryo testing. In a US sample (N = 1,467), participants were more willing to test for medical conditions (for example, heart disease) than non-medical traits (for example, antisocial behaviour or low intelligence), although over half supported testing for non-medical traits. In a forced-choice implantation task, participants used medical and non-medical information to a similar extent when making decisions. Similar patterns were observed in Chinese participants (N = 623). These findings suggest variability in choices across specific traits and conditions, rather than a uniform distinction between medical conditions and non-medical traits. The Stage 1 protocol for this Registered Report was accepted in principle on 7 October 2025. The protocol, as accepted by the journal, can be found at https://osf.io/vb9c2 .

Read article →

Refining the role of behavioural and social sciences in pandemic response post-COVID-19

2026-08-25 · DOI 10.1038/S41562-026-02506-4

Maja Graso; Karl Aquino; Stefan Baral; Nathan Berg; Cory J. Clark; Lucian Gideon Conway; Paul Dolan; Steven Grover; Kyriaki Fousiani; Claudia C. Kitz; Dario Krpan

Abstract. Abstract unavailable.

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Large language models do not have emotions

2026-08-24 · DOI 10.1038/S41562-026-02558-6

Amit Goldenberg; James J. Gross

Abstract. Abstract unavailable.

Read article →

Quantifying the prevalence and impact of overreaching causal claims in social science

2026-08-24 · DOI 10.1038/S41562-026-02553-X

Calvin Isch; Timothy Dorr; Neil Fasching; Grace Jennings; Duncan J. Watts

Abstract. Abstract Across the social sciences, many studies use cross-sectional designs that reveal associations but are generally unable to support direct causal claims, yet authors of such articles may make or imply causal claims anyway. Here, to examine the prevalence of such 'overreaching' causal language, we analysed 194,631 cross-sectional articles using large language models. Over the period 1980-2024, an average of 46% of articles contained causal language in their titles or

abstracts, where the annual rate has risen almost threefold since 2000 from 20% to 60%. To examine the effects of such language, we conducted a human-subjects experiment (N = 1, 105), finding that readers frequently indicate abstracts with this phrasing provide causal evidence but that methodological labels (= 0.4, 95% confidence interval 0.56 to 0.19) and associational wording (= 0.3, 95% confidence interval 0.43 to 0.07) reduce this tendency. Experiments with five LLMs revealed that model summaries of these articles (N = 100 each) can amplify causal overstatement, removing hedges and introducing causal claims where articles used strictly associational phrasing; however, prompting caution diminishes this pattern.

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The shrinking landscape of linguistic diversity in the age of large language models

2026-08-24 · DOI 10.1038/S41562-026-02550-0

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Abstract. Language is far more than a communication tool; it encodes a wealth of information about a person's identity, psychological state and social context, providing valuable insights for diverse fields including psychology, marketing and healthcare. Across three studies spanning seven datasets in different domains and over 880,000 texts, we show that the widespread adoption of large language models (LLMs) as writing assistants is linked to declines in linguistic diversity, interfering with the societal and psychological insights language provides. While core content is retained when LLMs polish and rewrite texts, LLMs also homogenize writing styles, reducing writing-complexity variance by a statistically significant 21-50% across datasets and models (P 0.05), and amplify patterns associated with dominant characteristics while suppressing others, emphasizing conformity over individuality. These trends hold across different LLMs, prompts and contexts, with potential implications for diagnostic processes, personalization efforts, hiring assessments and cultural preservation.

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Nature Communications

Navigating regulatory fragmentation in the convergence of synthetic biology, artificial intelligence, and automation

2026-08-29 · DOI 10.1038/S41467-026-77338-X

Ross K. J. McLennan; Paul S. Freemont; Isak S. Pretorius

Abstract. The convergence of synthetic biology, artificial intelligence (AI), and automation (SynBioxAI) creates research activities that simultaneously engage biosecurity, AI governance, export control, and data sovereignty frameworks - none of which are designed for convergent science. We conduct a comprehensive cross-jurisdictional analysis of this regulatory landscape across sixteen nations, identifying critical ambiguities where novel SynBioxAI objects fall between established regulatory categories. Through seven realistic collaboration scenarios, we demonstrate that regulatory friction is multiplicative rather than additive. We propose the SynBioxAI Regulatory Interoperability Toolkit (RIOT): a practical, seven-lens institutional framework that gives research institutions the capacity to navigate regulatory divergence efficiently and transparently. The convergence of synthetic biology, AI, and automation creates new challenges for regulatory frameworks. In this Perspective, the authors analyse national regulations and propose the SynBioxAI regulatory interoperability toolkit to give research institutes the capacity to navigate divergent regulatory landscapes.

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Agrivoltaics can reduce political polarization and local opposition to solar energy on land

2026-08-27 · DOI 10.1038/S41467-026-77141-8

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Abstract. Abstract As solar energy deployment on land intensifies, political polarization and local opposition increasingly challenge its social acceptance. We propose that agrivoltaics - a dual land-use technology that combines solar power generation with agricultural production - can help bridge ideological divides and reduce local opposition by minimizing land-use conflicts and offering additional benefits to rural communities. To test this hypothesis, we conducted a pre-registered, field-embedded experiment with a representative sample (2145 Swiss residents) just prior to a national renewable energy referendum, combining a personalized geolocated information treatment on local agrivoltaics potential with a conjoint experiment varying key technology and project design attributes. Here we show that agrivoltaics garners broad public support across ideological groups and reduces local opposition, especially when projects are well integrated into agricultural landscapes, do not significantly reduce agricultural yields, and are locally owned. These findings underscore the importance of technology and project design in addressing local and right-leaning opposition to land-based solar energy.

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A machine-assisted framework for systematic error analysis in clinical concept extraction

2026-08-26 · DOI 10.1038/S41467-026-77067-1

Sunyang Fu; Qiuhao Lu; Jaerong Ahn; Fang Chen; Hanyun Yin; Julia Wen; Zhiyi Yue; Taylor Harrison; Jiang Jun; Xiaoyang Ruan; Ming Huang; Andrew Wen; Liwei Wang; Min Ji Kwak; Nahid Rianon; Yanshan Wang; Ruihong Huang; Hongfang Liu

Abstract. Error analysis is critical for evaluating and improving clinical natural language processing (NLP) models, yet no standardized framework exists to support systematic error analysis for clinical concept extraction. Here we present MedError, a machine-assisted, human-in-the-loop framework integrating a validated error taxonomy, large language model (LLM)-assisted classification, and a structured annotation workflow. MedError is developed from 1187 unique errors manually curated from 4227 clinical notes across three institutions, spanning 25 error types and 48 clinical concept categories. Here we show that MedError supports both single-site and federated multisite error analysis through three complementary evaluations. First, benchmarking of six proprietary and open-source LLMs shows that automated error classification alone is insufficient without human involvement. Second, MedError-derived error feedback improves NLP extraction performance by 34-62% over an unstructured baseline across 11 cognitive status concepts. Third, prospective external validation at two independent institutions demonstrates improved agreement with expert-adjudicated annotations by up to 30% (McNemar exact test, p = 0.035, = 0.38) and reduced annotation time by 26-33%, providing a systematic and practical framework for real-world clinical NLP deployment. MedError, a machine-assisted human-in-the-loop framework, improves the standardization, validity, and efficiency of clinical NLP error analysis, supporting more transparent and trustworthy model deployment.

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Multi-source agent-based modeling to optimize influenza mitigation strategies in Hong Kong

2026-08-25 · DOI 10.1038/S41467-026-74857-5

Liping Peng; Yiyang Guo; Nicole N. Y. Tsang; Xiaotong Huang; Hau Chi So; Benjamin J. Cowling; Dennis K. M. Ip; Tim K. Tsang

Abstract. Seasonal influenza control faces challenges from variable vaccine effectiveness and uncertain non-pharmaceutical intervention (NPI) performance. While vaccination remains the primary strategy, its effectiveness varies between vaccine-matched and mismatched seasons. Increased post-COVID-19 NPI acceptance enhances influenza control feasibility, yet optimal combination approaches remain poorly understood. We use an agent-based model to analyze influenza transmission across six seasons in Hong Kong (2009-2013) with varying epidemic characteristics. We integrate surveillance, serological, and school absenteeism data for calibration, enabling accurate estimation of reported and unreported infections. We evaluate age-targeted vaccination, staying home when sick, mask use, and school-based interventions across diverse real-world scenarios. Compared to baseline, child vaccination consistently outperforms other strategies, with targeting those under 12 yielding the greatest population-level attack rate reduction (up to 8.5% relative reduction per 100,000 vaccinated). Among NPIs, 40% mask coverage reduces attack rates by 18%-43%, comparable to 25% of symptomatic individuals staying home. During vaccine-mismatched seasons, combining high-coverage mask use and staying home reduces attack rates by

79%-84%. High-coverage school-based vaccination is more effective than closures, reducing student attack rates by up to 85% versus 33% for 14-day closures. Our multi-source calibration approach provides robust evidence for prioritizing child vaccination and strategic NPI combinations. Modeling six influenza seasons to evaluate mitigation strategies, this study proposes a signal-responsive policy framework. Findings show child vaccination and mask use curb transmission; their combination is vital during vaccine mismatches.

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Nature Machine Intelligence

The epistemic debt of generative AI

2026-08-26 · DOI 10.1038/S42256-026-01294-W

Irene Unceta; Paula Subias-Beltran; Oriol Pujol

Abstract. Abstract unavailable.

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Late Additions

Proceedings of the National Academy of Sciences

Group size effects and collective misalignment in LLM multi-agent systems

2026-08-18 · DOI 10.1073/PNAS.2531697123

Ariel Flint; Luca Maria Aiello; Romualdo Pastor-Satorras; Andrea Baronchelli

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Abstract. Multi-agent systems of large language models (LLMs) are rapidly expanding across domains, introducing dynamics not captured by single-agent evaluations. Yet, existing work has mostly contrasted the behavior of a single agent with that of a collective of fixed size, leaving open a central question: How does group size shape dynamics? Here, we move beyond this dichotomy and systematically explore outcomes across the full range of group sizes. We focus on multi-agent misalignment, building on recent evidence that interacting LLMs playing a simple coordination game can generate collective biases absent in individual models. First, we show that collective bias is a deeper phenomenon than previously assessed: Interaction can amplify individual biases, introduce new ones, or override model-level preferences. Second, we demonstrate that group size affects the dynamics in a nonlinear way, revealing model-dependent dynamical regimes. Finally, we develop a mean-field analytical approach and show that, above a critical population size, simulations converge to deterministic predictions that expose the basins of attraction of competing equilibria. These findings establish group size as a key driver of multi-agent dynamics and highlight the need to consider population-level effects when deploying LLM-based systems at scale.

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Scientific production in the era of large language models: Outcome-triggered treatment timing and spurious event-study dynamics

2026-08-11 · DOI 10.1073/PNAS.2618638123

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Abstract. Large language models (LLMs) are increasingly used in scientific writing, but their effect on individual productivity is difficult to identify because adoption is rarely directly observed. [K. Kusumegi et al. , Science 390 , 1240-1243 (2025)] infer adoption from the first paper detected as LLM-assisted and report large productivity gains after adoption. We show that this treatment-timing rule mechanically generates positive event-study dynamics even in the absence of any causal effect. Because high-output months are more likely to produce a detected paper, treatment assignment becomes intrinsically linked to productivity. Using reconstructed arXiv data, we show that random treatment assignments, neutral-keyword triggers, inverted treatment, and pre-ChatGPT placebo periods all generate similar dynamics. Simulations with no treatment effect also reproduce the same posttreatment patterns reported in K. Kusumegi et al. , Science 390 , 1240-1243 (2025). Our results demonstrate that first-detection timing alone creates spurious evidence of productivity gains from LLM adoption.

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Nature Machine Intelligence

Towards principled knowledge editing methods for large language model reasoning

2026-08-14 · DOI 10.1038/S42256-026-01276-Y

Ningyu Zhang; Yunzhi Yao; Jiaxin Qin; Haoming Xu; Yuqi Zhu; Zeping Yu; Mengru Wang; Yuqi Tang; Jia-Chen Gu; Shumin Deng; Huajun Chen

Abstract. Abstract unavailable.

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